

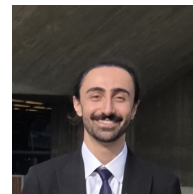
Kasra Mohammadi

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ABOUT ME

I am a mechanical engineer with an MSc in Sustainable Energy Technology from TU Delft, specialising in heating and cooling systems, thermal energy storage, and energy system modelling. I work on thermal energy systems and grid congestion at the Ingenieursbureau of Gemeente Rotterdam. I am experienced in developing Python-based engineering models and analysing ATES (WKO), building energy, and renewable energy systems, and am particularly interested in sustainable heating and cooling, energy infrastructure, and the energy transition.

EDUCATION

Delft University of Technology (TU Delft)

M.Sc. Sustainable Energy Technology, **cum laude** | GPA: **8.96/10**

Delft, Netherlands

September 2024 – August 2026

- Specialisation: Heating and Cooling.
- Thesis (grade 9/10): *Mitigating electricity grid congestion with medium-temperature aquifer thermal energy storage* (TU Delft repository).

University of Tehran (UT)

B.Sc. Mechanical Engineering | GPA: **19.18/20**

Tehran, Iran

September 2019 – February 2024

WORK EXPERIENCE

Generation

Trainee Technisch Management – Gemeente Rotterdam, Ingenieursbureau

Rotterdam, Netherlands

October 2026 – present

- Two-year traineeship in integral technical management for public-sector infrastructure, specialising in the technical management (IPM) role.
- Working on thermal energy systems and grid congestion at the municipal engineering bureau.

Gemeente Rotterdam – Ingenieursbureau

Graduation Project Intern – Thermal Systems and Grid Congestion

Rotterdam, Netherlands

January 2026 – August 2026

- Conducted MSc thesis research on grid congestion mitigation using medium-temperature aquifer thermal energy storage (MT-ATES).
- Used Python, SEAWAT/MODFLOW, and TESPpy to assess the potential of MT-ATES to reduce winter peak electricity demand.

Techniplan Adviseurs BV

Intern

Rotterdam, Netherlands

September 2025 – December 2025

- Developed a workflow based on SEAWAT, Python, and Excel to simulate subsurface storage conditions of ATES systems.
- Made the company's ATES modelling procedure more flexible, automated, and user-friendly.

Advanced Energy Systems Lab (AESL) – University of Tehran

Research Assistant

Tehran, Iran

June 2023 – July 2024

- Worked on the design and analysis of a near-zero-energy primary school building with PV integration across different Iranian climates.
- The project increased the energy independence of schools from the electricity grid in underdeveloped regions.
- Used TRNSYS, EnergyPlus, and MATLAB to obtain, validate, and evaluate the results.

Department of Geoinformation – AGH University

Paid Research Intern

Kraków, Poland

Summer 2022

- Performed laboratory tests, processed experimental data, and supported analysis for a wastewater treatment research project.
- Assisted in onboarding and guiding another international intern in laboratory procedures and project tasks.

VOLUNTEERING

Energy for Refugees (EfR)

Board Member and Engineer

Delft, Netherlands

January 2025 – January 2026

- Designed and sized a self-sustaining PV-battery system for a vocational school in Kakuma refugee camp, Kenya.
- Estimated demand profiles, optimized system sizing under funding constraints.
- The project resulted in a 58.5 kWp solar installation currently installed and operational.

Delft Sustainable Energy Association (DelftSEA)

Board Member and Treasurer

Delft, Netherlands

November 2024 – November 2025

- Managed the association's finances, budgeting, and cash flow to support the association in achieving its goals.
- Helped with the organization of student events, improving the social and academic environment.

University of Tehran Emergency Board of Studies

Member

Tehran, Iran

September 2022 – September 2023

- Member of the Emergency Board of Studies formed during the 2022 national student protests in Iran.
- My role as a member was to conduct meetings with professors and academic staff to prevent hindrances to the educational quality of the course and help students express their demands during the protests.

ACADEMIC EXPERIENCE

Teaching Assistant – TU Delft and University of Tehran**September 2020 – February 2026**

- Teaching assistance for courses with a combined study load of over 70 ECTS.
- Responsibilities included tutoring, recitation classes, assignment and quiz design, grading, and guiding students during projects.
- Course in TU Delft: SET3061 – Energy System Modelling
- Courses in University of Tehran: Optimization of Mechanical Systems, Thermodynamics I, Fluid Mechanics I, Physics I, Numerical Calculations

SKILLS

Programming and Modelling: Python, MODFLOW, SEAWAT, FloPy, MATLAB, C++, EES**Energy and Simulation Tools:** TRNSYS, EnergyPlus, TESPpy, CoolProp, WANDA, COMSOL**Engineering Software:** SolidWorks, ANSYS Fluent, MSC ADAMS**Interests:** Photography, history, learning languages, guitar (mainly metal), blues harmonica

LANGUAGE SKILLS

- Persian: Native
- English: Fluent (C2)
- Turkish: Fluent (C2)
- Dutch: Intermediate (B1)
- Italian: Intermediate (B1)

AWARDS AND HONOURS

- Holder of the “National Examination Top Rank Prize” scholarship awarded by the Fanni Foundation, 2019–2023
- Ranked **46th** among ~160,000 participants in the 2019 National Iranian University Entrance Exam (Mathematics and Physics Field)
- Ranked **5th** in the nation among ~150,000 participants in the 2019 National Iranian University Entrance Exam (Foreign Languages Field)
- Member of Iran's National Elites Foundation (INEF), 2019 – 2024

REFERENCES

References available upon request.